

Robotics education

Robhatah and PurpleLeap have tied up to provide high quality robotics education in India

Electronic skin

Using nano-scale transistors and rubber, Stanford University researchers have created an electronic skin – an artificial layer with tactile sensitivity – that could replace touch-sensors pretty soon, in robots

Feature

4. **Virtual robots:** Also known as bots or web-bots, these are actually software programmes that simulate real robots or function like human beings in the virtual world. WebCrawlers and ELIZA – the chatterbot that simulates conversation between two people on the web – are examples of virtual bots.
5. **Shape-shifting robots:** Recently, platforms for robots have moved towards polymorphism. Serpentine robots are the best example of shape-changing robots. Recently, Carnegie Mellon University's Biorobotics Lab developed a snake-like modular robot called Uncle Sam that looks like a series of tin cans and climbs poles and trees before it surveys the landscape with a camera on its head.
6. **Swarm robots:** Inspired by the teamwork of insects like ants, scientists have been working on creating a large number of little robots working on a single task – multi-agent teams that

TETE-A-TETE WITH ABHEEK BOSE, FOUNDER, ROBOTS ALIVE



Mr. Abheek Bose, founder Robots Alive

Are the entrepreneurial efforts in the field of robotics succeeding in India?

Starting up in the robotics domain is still a struggle. There is a significant lack of skill and know-how as far as robotics is concerned so it is very difficult to get the correct kind of people in the job.

Is the Indian education system conducive to robotics?

The educational system in India is not conducive to Robotics, because it is still considered a subject instead of a stream of engineering. In developed nations, robotics is taught as an engineering discipline.

What about government policies?

Robotics is still very new in India and the government is still trying to get the relevant policies in space.

What does the future hold, over the next 50 years?

There are already successful robots which perform small scale domestic tasks like vacuum cleaning (Roomba & Scooba from iRobot). There is also large scale research especially in US, Japan, Korea and Germany which deal with fully capable house-keeping robots. Further, there is a worldwide competition of house-keeping robotics, Robocup at Home.

In the next 50 years, robots will be most definitely be doing some of mankind's everyday domestic chores. I am still doubtful, however, that we will see this in India over the next 50 years...

the server is the location of the ball.

Given the rapid developments in the field of robotics, what can we expect in the future? According to Dr. Vadakkepat and FIRA President Dr. Kim Jong-Hwan, "we, as humans have not been able to adopt robots into our daily lives because we are emotional beings. Aibo, Sony's robot pet for example, wasn't adopted as popularly for the lack of association in a more realistic nature. You could tap and pet it, but you couldn't get it to act naturally stupid!"

Why, robot?

Robots are also finding their way into fields such as surgery and medicine.

During an important operation, where human errors can occur, perhaps the future really is robot surgeons that are overseen by human doctors. It might scare you to know that a robot is operating on you, but it's no scarier than worrying about whether your human surgeon has slept, isn't in a bad mood or under stress.

In the automotive industry, when your car is being painted, you can be sure that the robot painter has used the optimum amount of paint with an evenness of the coat not achievable by human hands.

When building or repairing complex structures such as suspension

bridges, which currently require humans to risk life and limb to get to the areas in need of repair, a robot could do the job without any nervousness.

However, all said and done, perhaps we're expecting too much too soon from the robotics field. Yes we can't have our own Bicentennial Man in our lifetimes, but we're certainly going to see robots enter our lives. However, in order for our country to stay amongst the top runners of this exciting and promising segment of technology, there needs to be youth involvement. Watch out for our December 2010 issue, when we will bring you closer to the world of robotics. **d**

ASUS VG236H 3D LCD MONITOR

IMMERSE YOURSELF IN
ULTRA-SMOOTH AND VIVID 3D ACTION

ASUS - No.1 in Quality and Services - The Wall Street Journal Asia

www.asus.in | Toll Free No. 1800-2090-365 | Email: reachus@asus.com

LCD MONITOR
VG236H

3D VISION
READY

ASUS
Inspiring Innovation • Persistent Perfection

120Hz
Double-frame Rate

2ms
Trace Free

Full HD 1080

HDMI

COLOR Shine
Technology

Splendid

Height Adjustment

JUST CAUSE 2